# הנחיות למצבי החייאה

# AMERICN HEART ASSOCIATION

## International Consensus on Cardiopulmonary Resuscitation and Emergency Cardiovascular Care

*Circulation*
*Volume 132(18 suppl 2)*
*November 3, 2015*

*Circulation*
*Volume 122(18 suppl 3)*
*November 2, 2010*

American Heart Association

נבחנים יקרים,

מצגת זו מרכזת עבורכם את ההנחיות להחייאה וניהול מצבי חירום לבביים, כפי שפורסמו ע"י American Heart Association בשנים 2010, 2015 (בהנחיות 2015 עודכנו רק חלק מהאלגוריתמים).

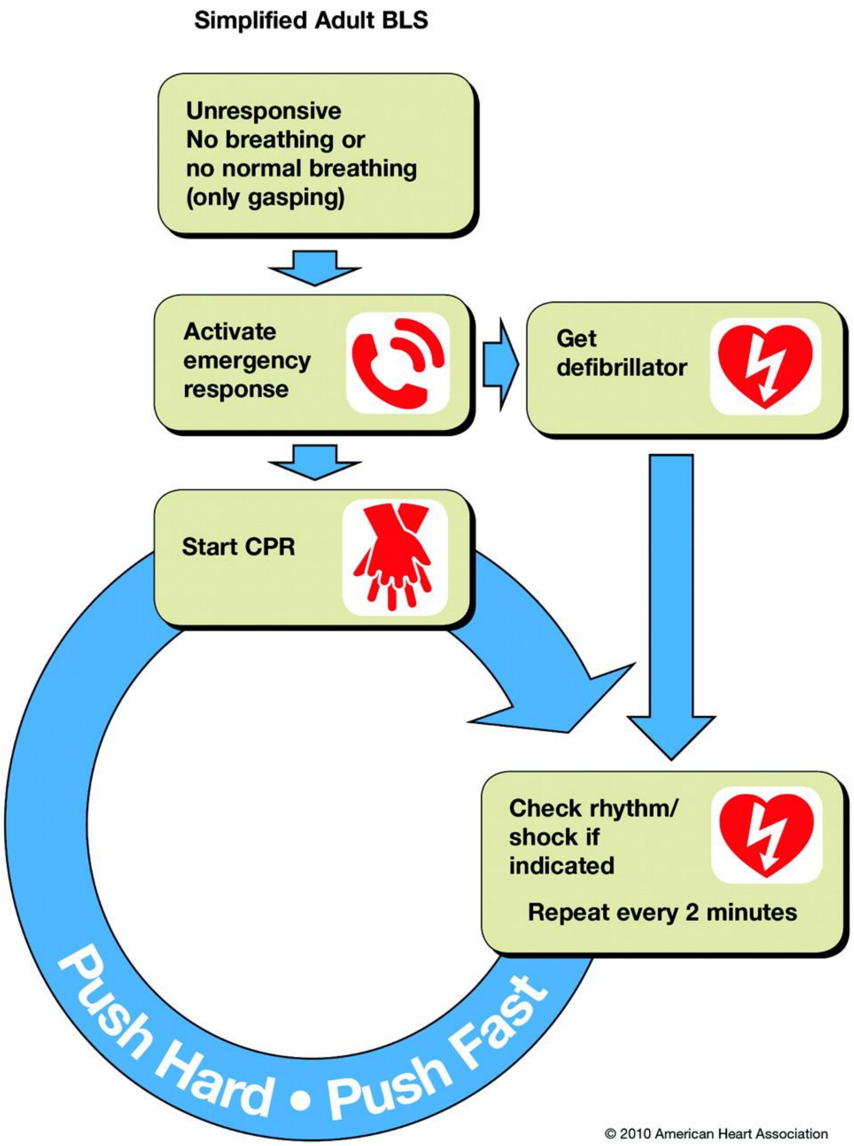
מצבי חירום נוספים (מצוקה נשימתית לסוגיה, בטן חריפה, דהידרציה, כוויות, נשיכות בע"ח ומצבים רבים אחרים) מוצגים במקומות אחרים ברשימת הספרות הרשמית לבחינה, בפרקים הרלבנטיים של ספרי הלימוד הרשמיים לגבי אותו מצב חירום, או בהנחיות קליניות רלבנטיות לנושא.

הנבחן נדרש לדעת לנהל את כל מצבי החירום שאותם ניתן לפגוש במרפאה!

בברכת בהצלחה,

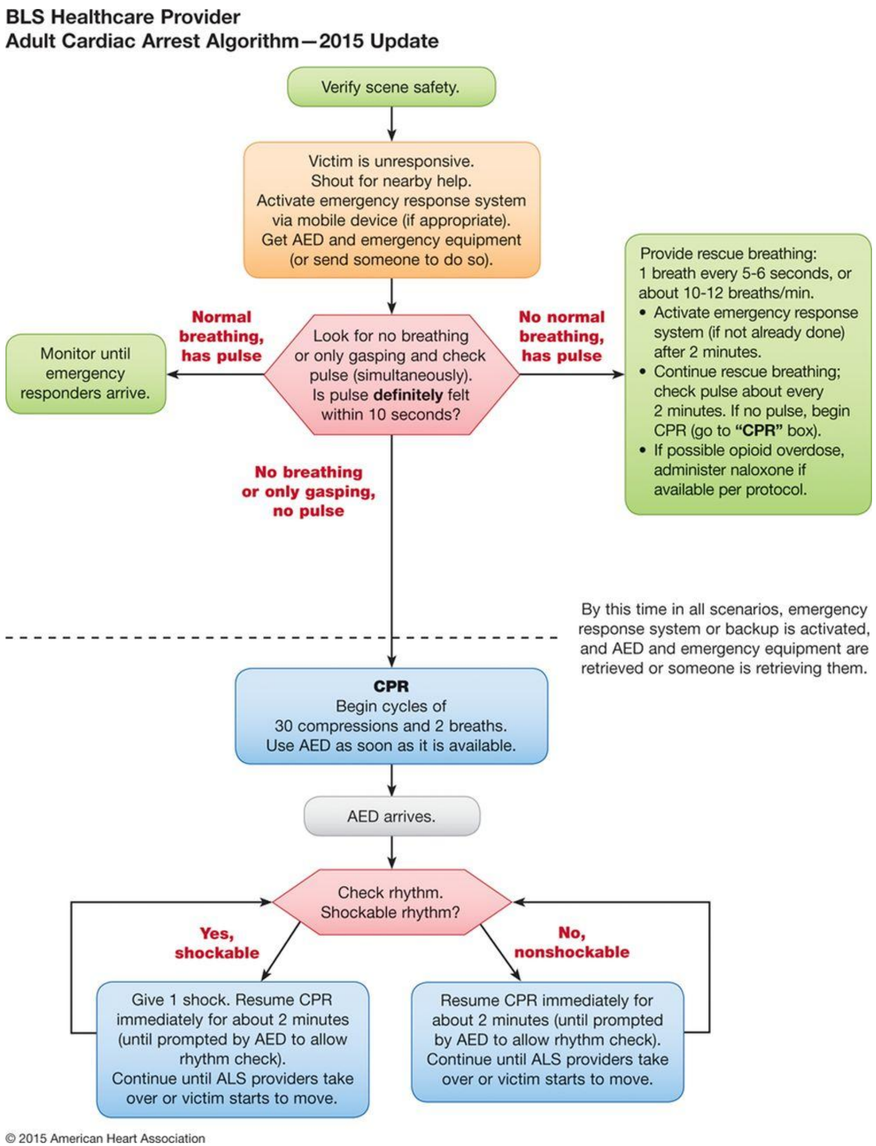
ועדת הבחינות ברפואת המשפחה

## Simplified Adult BLS Algorithm.



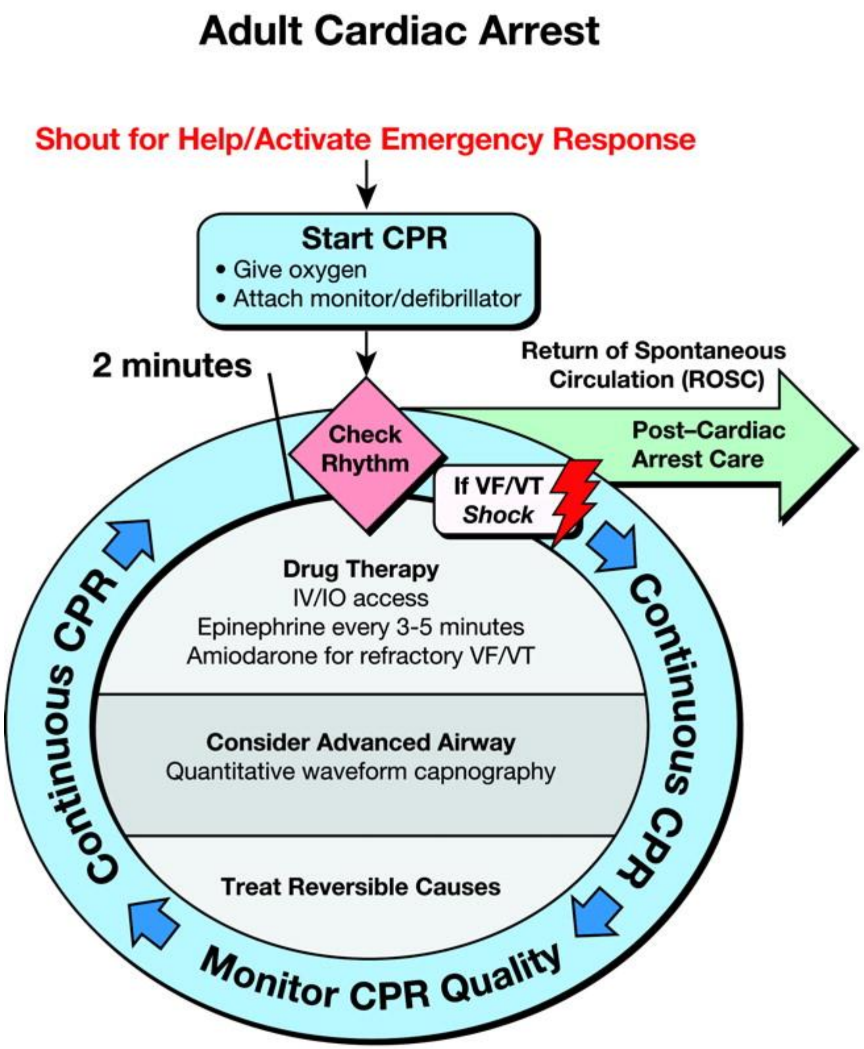

Andrew H. Travers et al. Circulation. 2010;122:S676-S684

# BLS Healthcare Provider Adult Cardiac Arrest Algorithm—2015 Update.




Monica E. Kleinman et al. Circulation. 2015;132:S414-S435

# ACLS Cardiac Arrest Circular Algorithm.






**CPR Quality**
- Push hard (≥2 inches [5 cm]) and fast (≥100/min) and allow complete chest recoil
- Minimize interruptions in compressions
- Avoid excessive ventilation
- Rotate compressor every 2 minutes
- If no advanced airway, 30:2 compression-ventilation ratio
- Quantitative waveform capnography
  - If $P_{ETCO_2}$ <10 mm Hg, attempt to improve CPR quality
- Intra-arterial pressure
  - If relaxation phase (diastolic) pressure <20 mm Hg, attempt to improve CPR quality

**Return of Spontaneous Circulation (ROSC)**
- Pulse and blood pressure
- Abrupt sustained increase in $P_{ETCO_2}$ (typically ≥40 mm Hg)
- Spontaneous arterial pressure waves with intra-arterial monitoring

**Shock Energy**
- **Biphasic:** Manufacturer recommendation (eg, initial dose of 120-200 J); if unknown, use maximum available. Second and subsequent doses should be equivalent, and higher doses may be considered.
- **Monophasic:** 360 J

**Drug Therapy**
- **Epinephrine IV/IO Dose:** 1 mg every 3-5 minutes
- **Vasopressin IV/IO Dose:** 40 units can replace first or second dose of epinephrine
- **Amiodarone IV/IO Dose:** First dose: 300 mg bolus. Second dose: 150 mg.

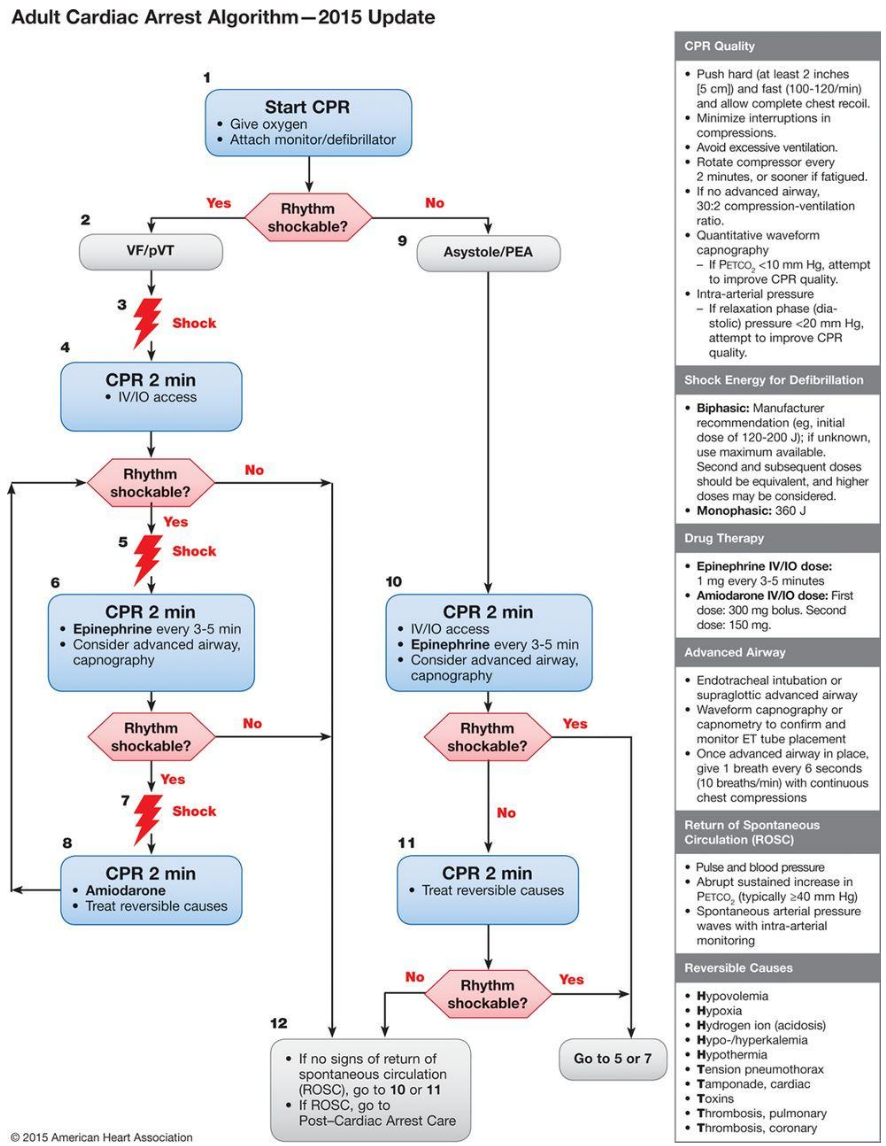
**Advanced Airway**
- Supraglottic advanced airway or endotracheal intubation
- Waveform capnography to confirm and monitor ET tube placement
- 8-10 breaths per minute with continuous chest compressions

**Reversible Causes**
- **H**ypovolemia
- **H**ypoxia
- **H**ydrogen ion (acidosis)
- **H**ypo-/hyperkalemia
- **H**ypothermia
- **T**ension pneumothorax
- **T**amponade, cardiac
- **T**oxins
- **T**hrombosis, pulmonary
- **T**hrombosis, coronary

**Robert W. Neumar et al. Circulation. 2010;122:S729-S767**

American Heart Association.

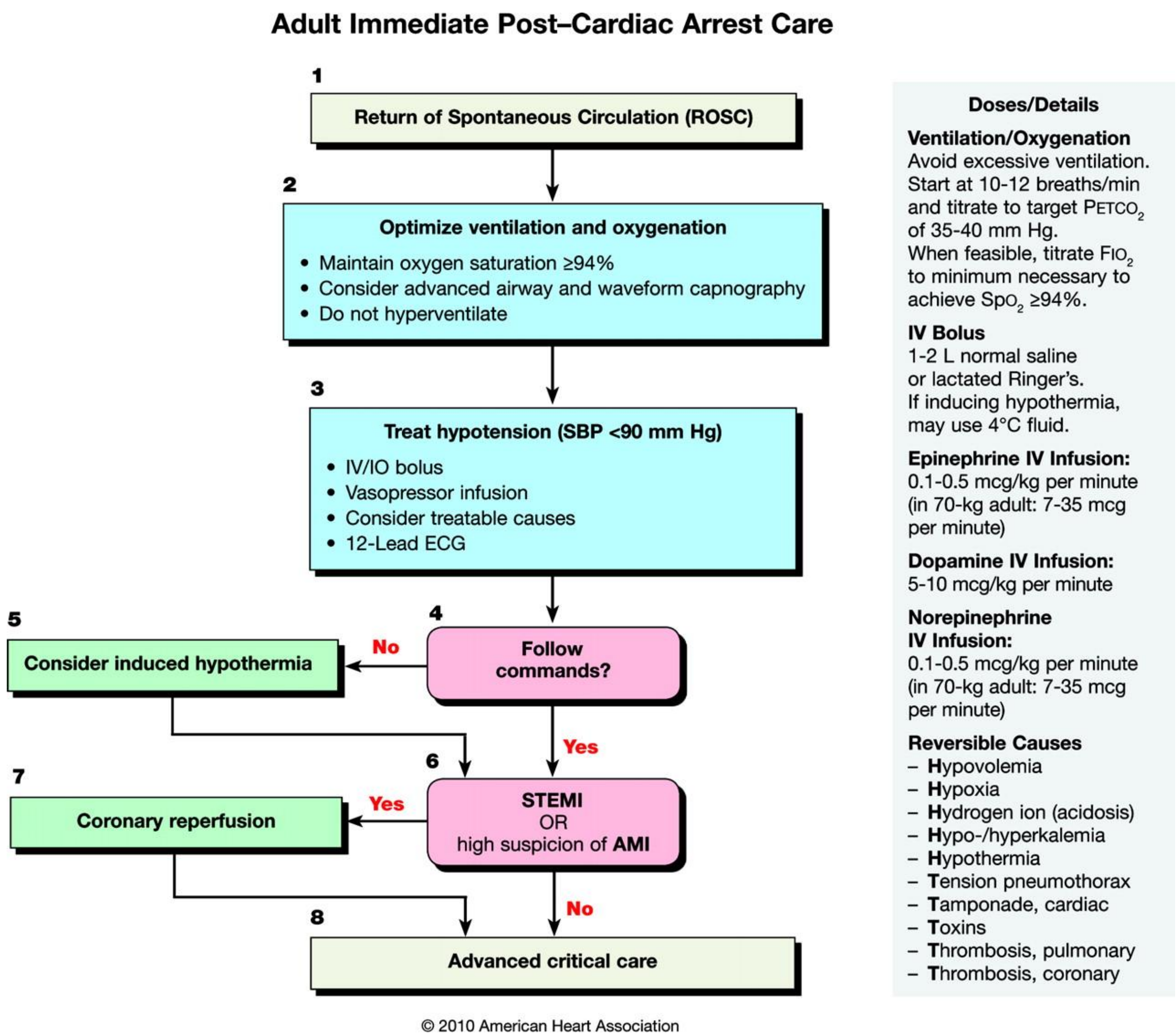
# Adult Cardiac Arrest Algorithm—2015 Update.




Mark S. Link et al. Circulation. 2015;132:S444-S464

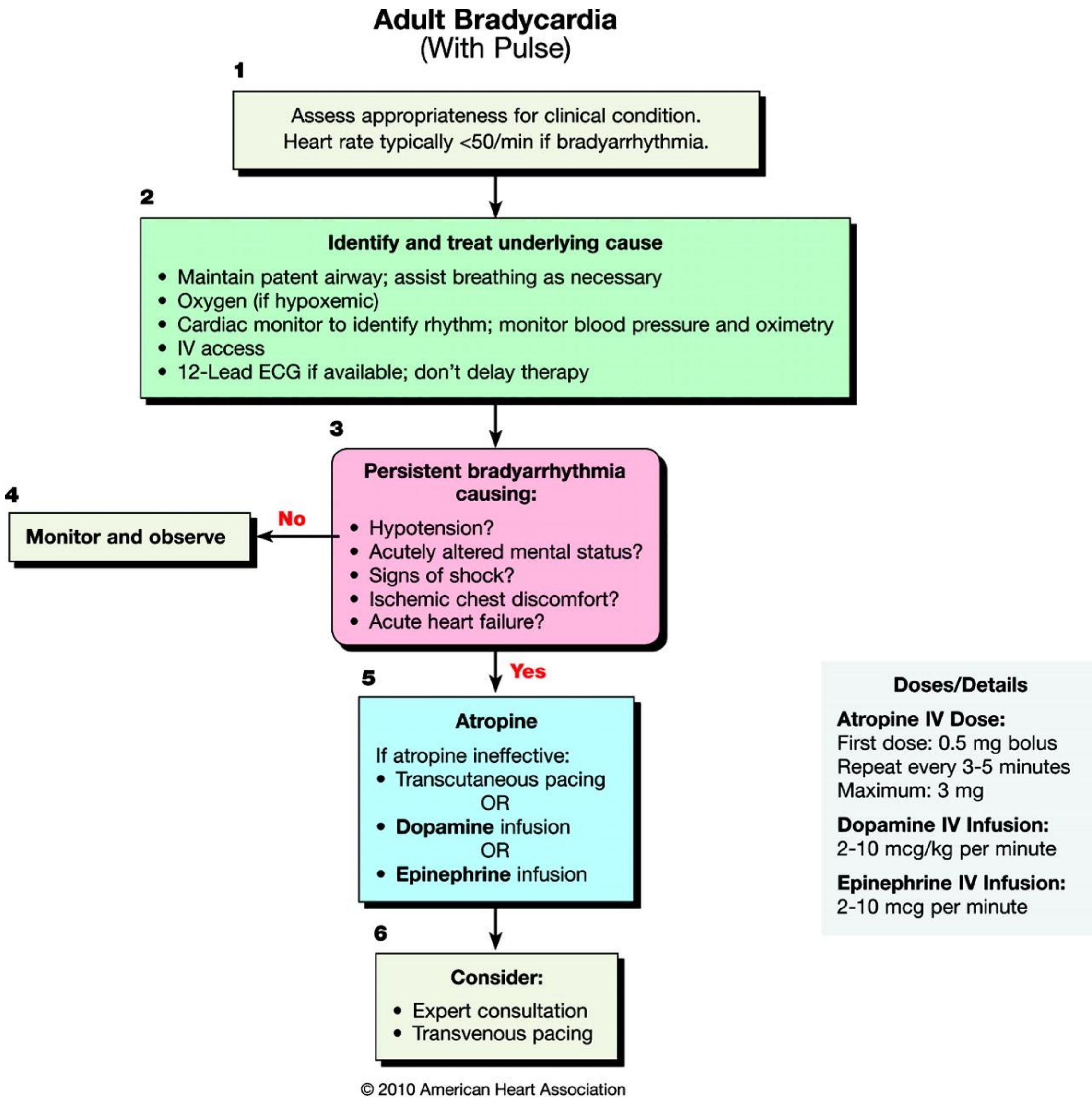
American Heart Association.

# Post–cardiac arrest care algorithm.



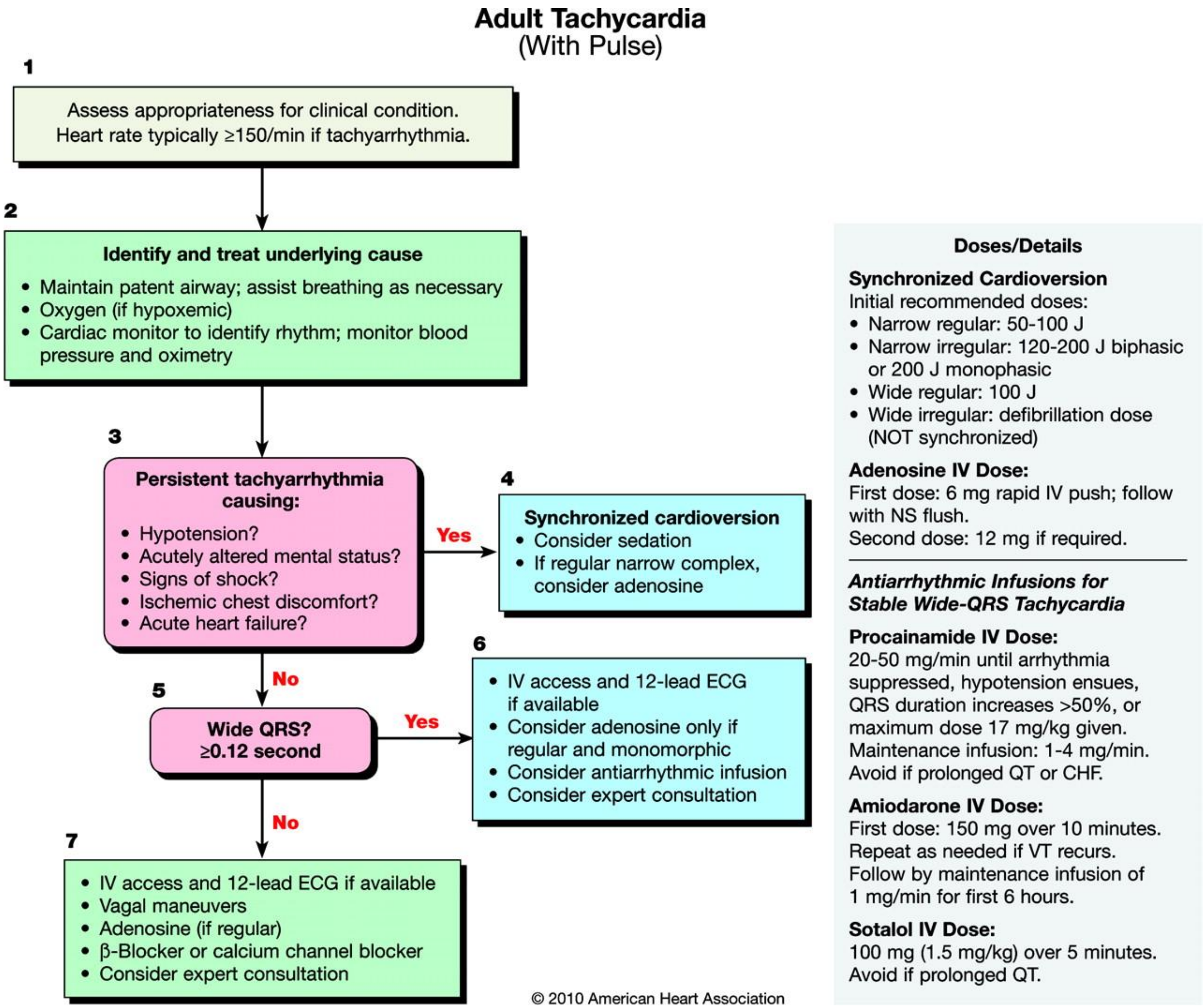

Mary Ann Peberdy et al. Circulation. 2010;122:S768-S786

# Bradycardia Algorithm.



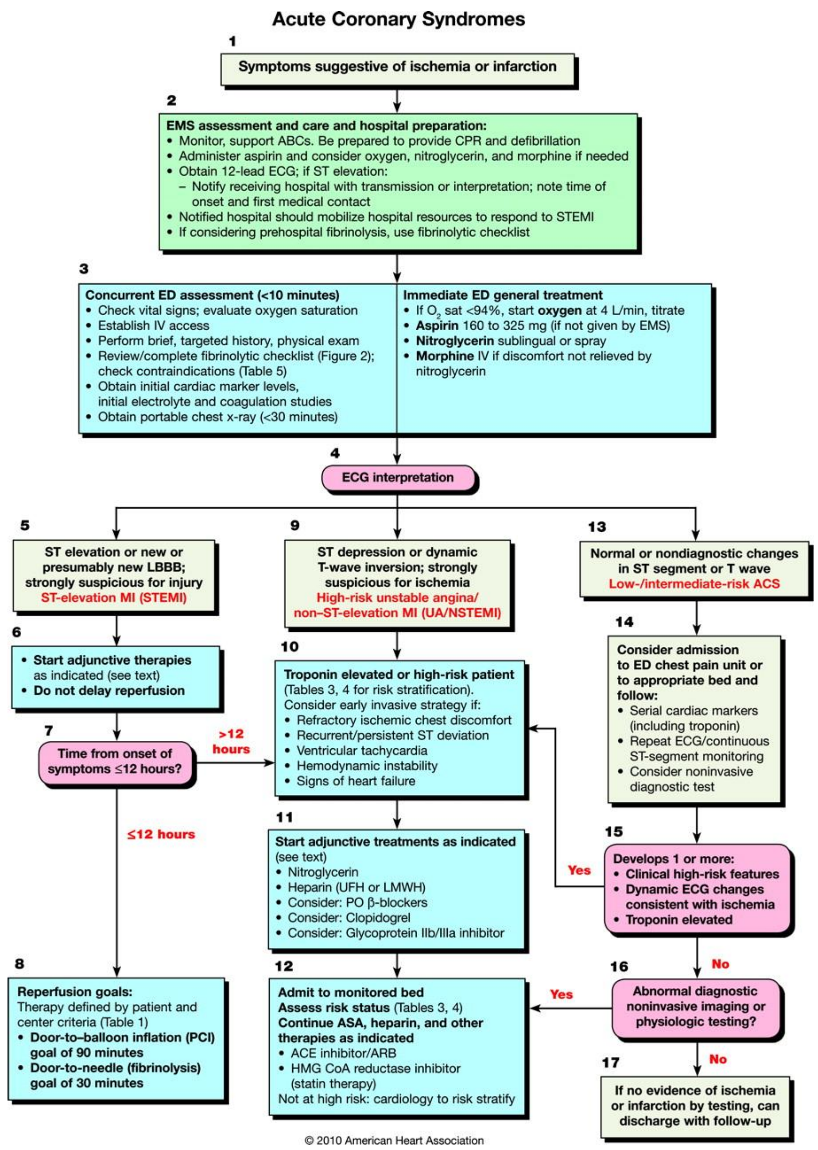

**Robert W. Neumar et al. Circulation. 2010;122:S729-S767**

# Tachycardia Algorithm.




**Robert W. Neumar et al. Circulation. 2010;122:S729-S767**

# Acute Coronary Syndromes Algorithm.




**Robert E. O'Connor et al. Circulation. 2010;122:S787-S817**

# Prehospital fibrinolytic checklist.

## Prehospital Fibrinolytic Checklist*

**Step 1** Has patient experienced chest discomfort for greater than 15 minutes and less than 12 hours?

YES → Does ECG show STEMI or new or presumably new LBBB?

NO → STOP

Does ECG show STEMI or new or presumably new LBBB?

YES → Step 2

NO → STOP

**Step 2** Are there contraindications to fibrinolysis?
If ANY one of the following is checked YES, fibrinolysis MAY be contraindicated.

| | | |
|---|---|---|
| Systolic BP >180 to 200 mm Hg or diastolic BP >100 to 110 mm Hg | ◯ YES | ◯ NO |
| Right vs left arm systolic BP difference >15 mm Hg | ◯ YES | ◯ NO |
| History of structural central nervous system disease | ◯ YES | ◯ NO |
| Significant closed head/facial trauma within the previous 3 months | ◯ YES | ◯ NO |
| Stroke >3 hours or <3 months | ◯ YES | ◯ NO |
| Recent (within 2-4 weeks) major trauma, surgery (including laser eye surgery), GI/GU bleed | ◯ YES | ◯ NO |
| Any history of intracranial hemorrhage | ◯ YES | ◯ NO |
| Bleeding, clotting problem, or blood thinners | ◯ YES | ◯ NO |
| Pregnant female | ◯ YES | ◯ NO |
| Serious systemic disease (eg, advanced cancer, severe liver or kidney disease) | ◯ YES | ◯ NO |

**Step 3** Is patient at high risk?
If ANY one of the following is checked YES, consider transfer to PCI facility.

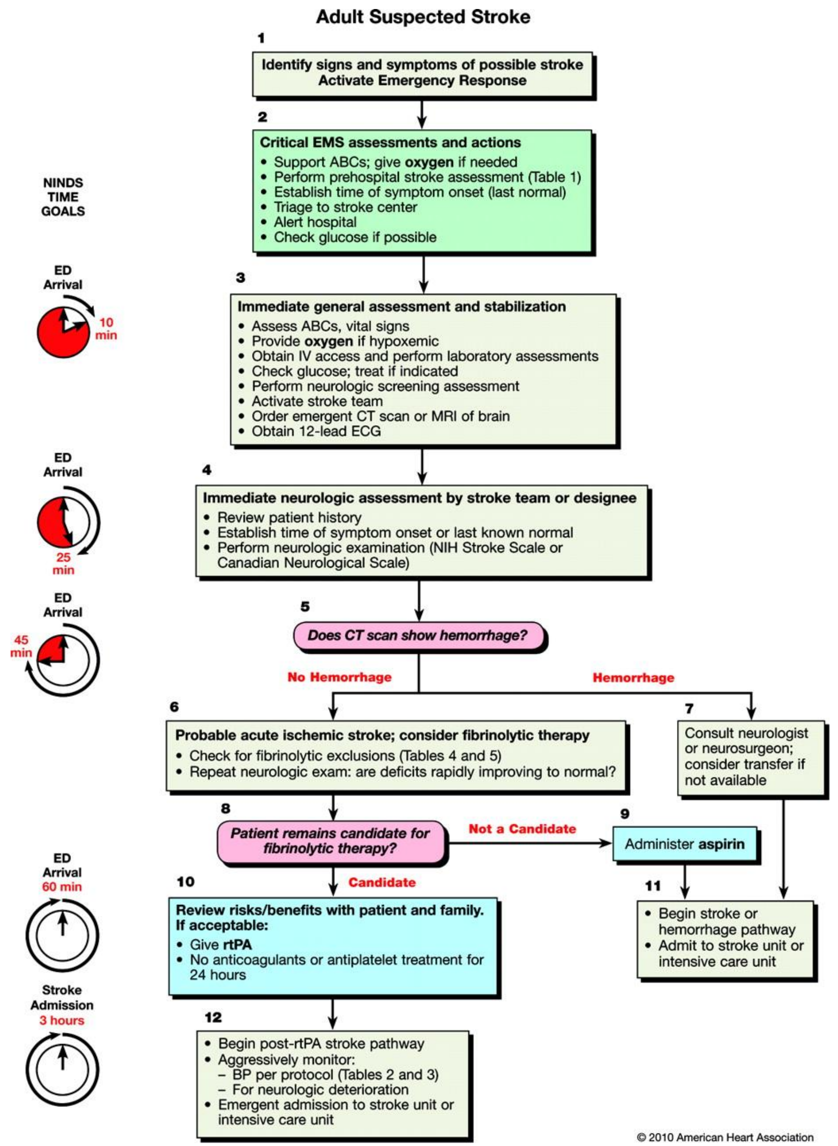
| | | |
|---|---|---|
| Heart rate ≥100/min AND systolic BP <100 mm Hg | ◯ YES | ◯ NO |
| Pulmonary edema (rales) | ◯ YES | ◯ NO |
| Signs of shock (cool, clammy) | ◯ YES | ◯ NO |
| Contraindications to fibrinolytic therapy | ◯ YES† | ◯ NO |
| Required CPR | ◯ YES | ◯ NO |

*Contraindications for fibrinolytic use in STEMI are viewed as advisory for clinical decision making and may not be all-inclusive or definitive. These contraindications are consistent with the 2004 ACC/AHA Guidelines for the Management of Patients With ST-Elevation Myocardial Infarction.

†Consider transport to primary PCI facility as destination hospital.

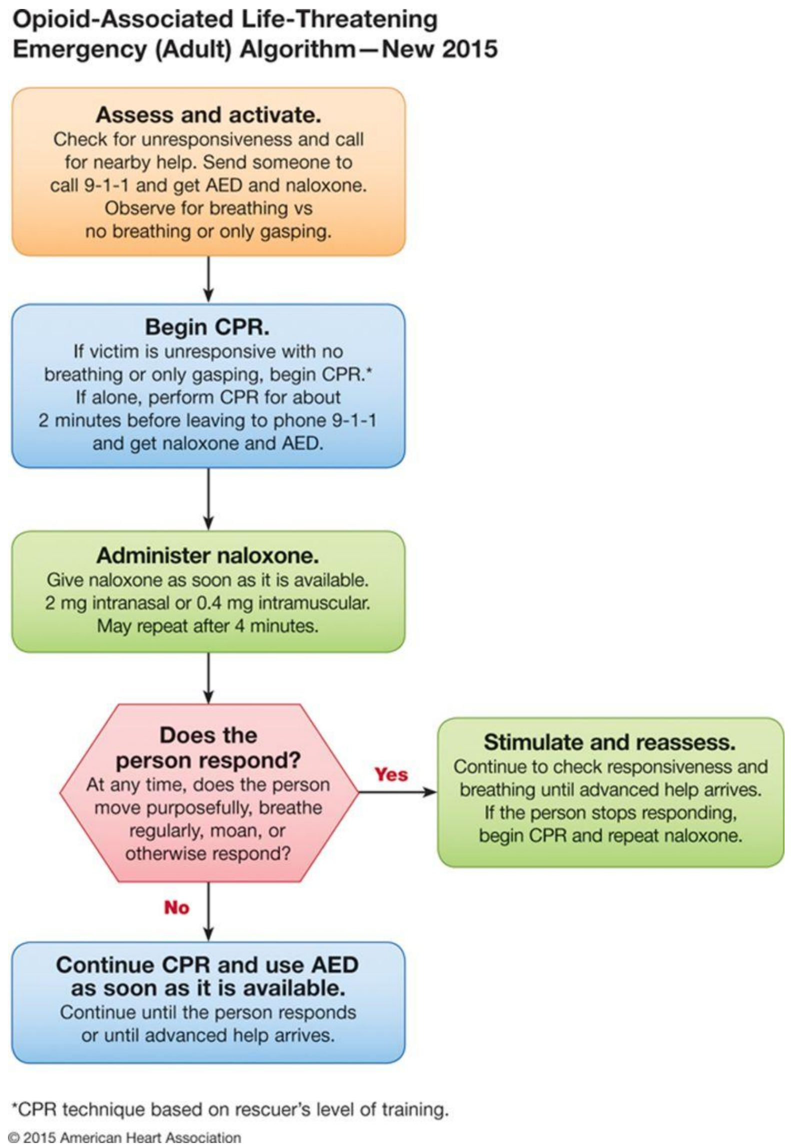
Robert E. O'Connor et al. Circulation. 2010;122:S787-S817

# Goals for management of patients with suspected stroke.




Edward C. Jauch et al. Circulation. 2010;122:S818-S828

# Opioid-Associated Life-Threatening Emergency (Adult) Algorithm.




Eric J. Lavonas et al. Circulation. 2015;132:S501-S518

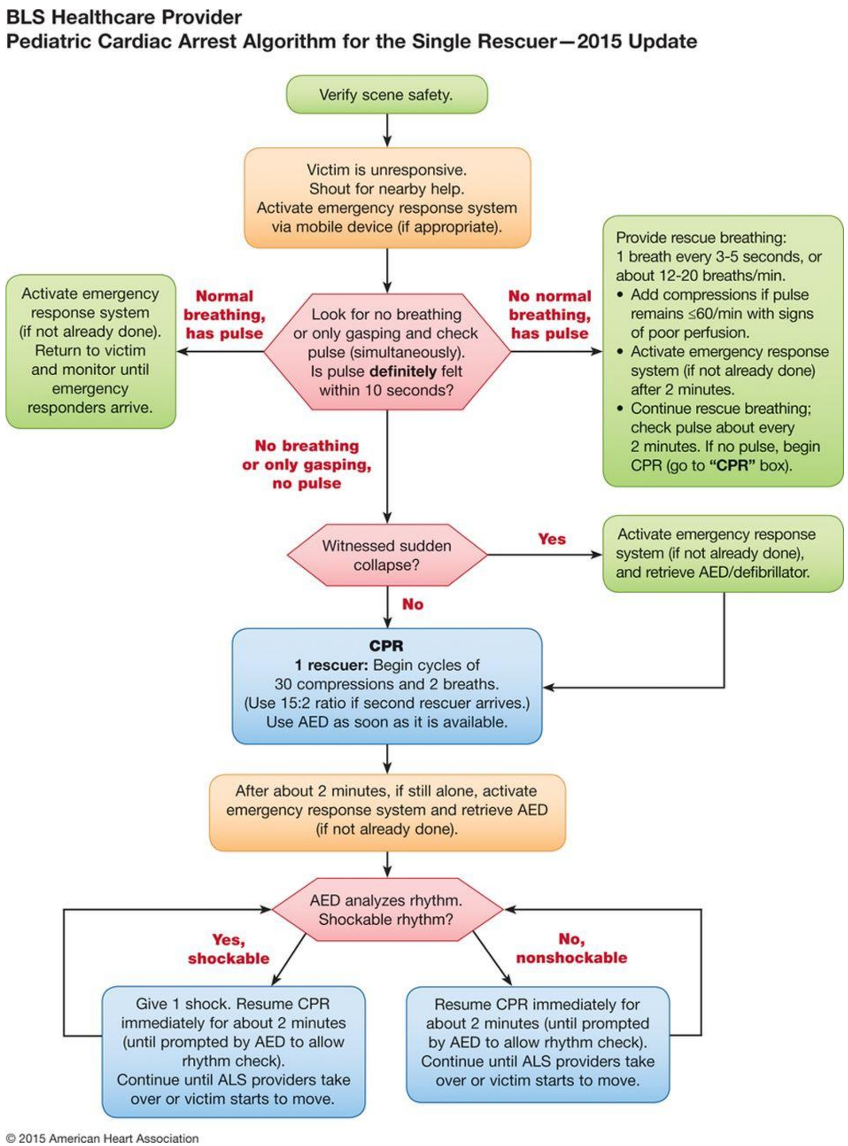
American Heart Association.

## Anaphylaxis

ראו הנחיות בקישור הבא מתוך ACLS 2005 (לא עודכן מאז)
http://circ.ahajournals.org/content/112/24_suppl/IV-143

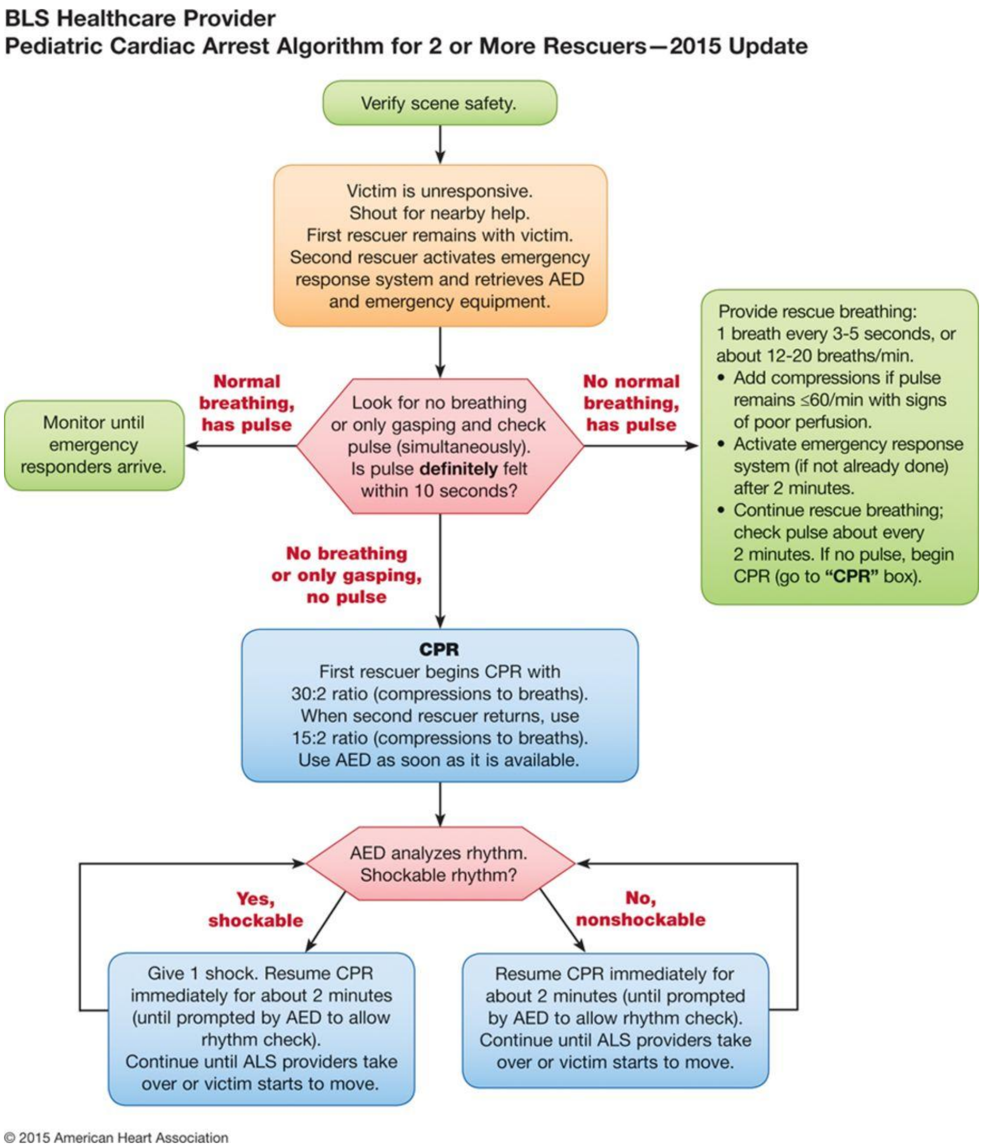
American Heart Association

# BLS Healthcare Provider Pediatric Cardiac Arrest Algorithm for the Single Rescuer—2015 Update.



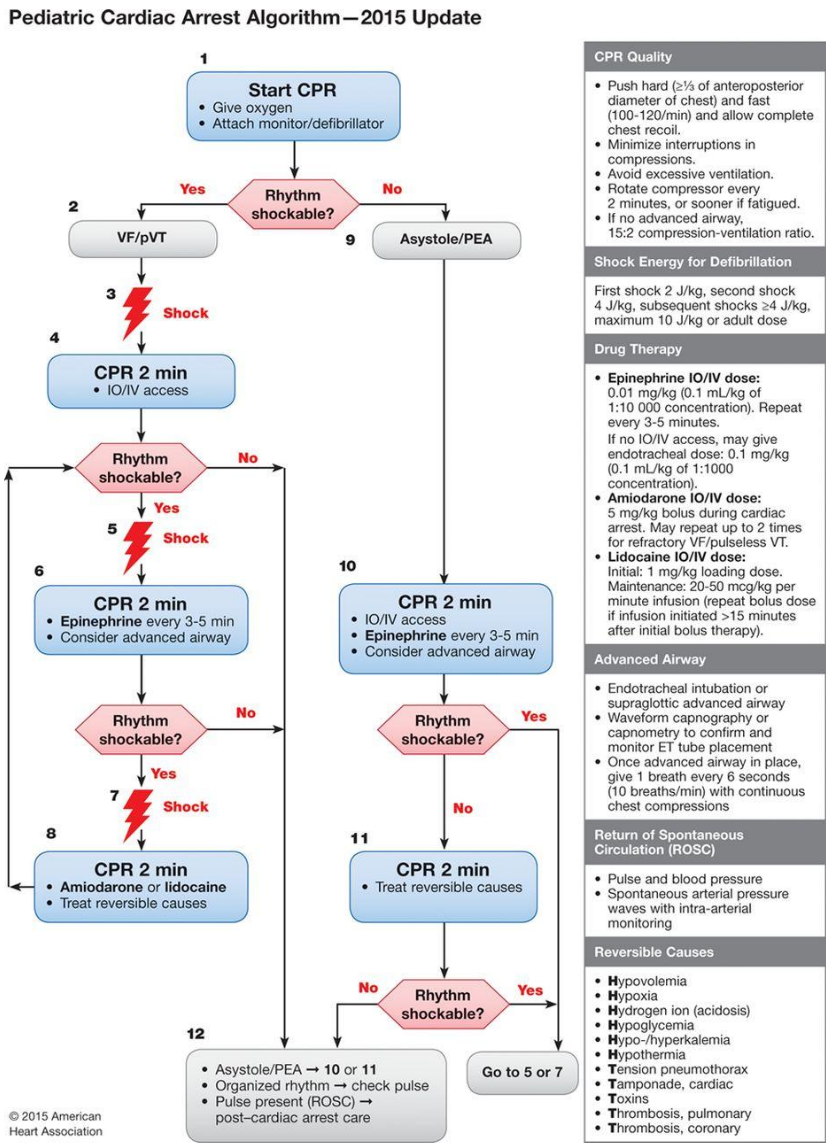

**Dianne L. Atkins et al. Circulation. 2015;132:S519-S525**

# BLS Healthcare Provider Pediatric Cardiac Arrest Algorithm for 2 or More Rescuers—2015 Update.




Dianne L. Atkins et al. Circulation. 2015;132:S519-S525

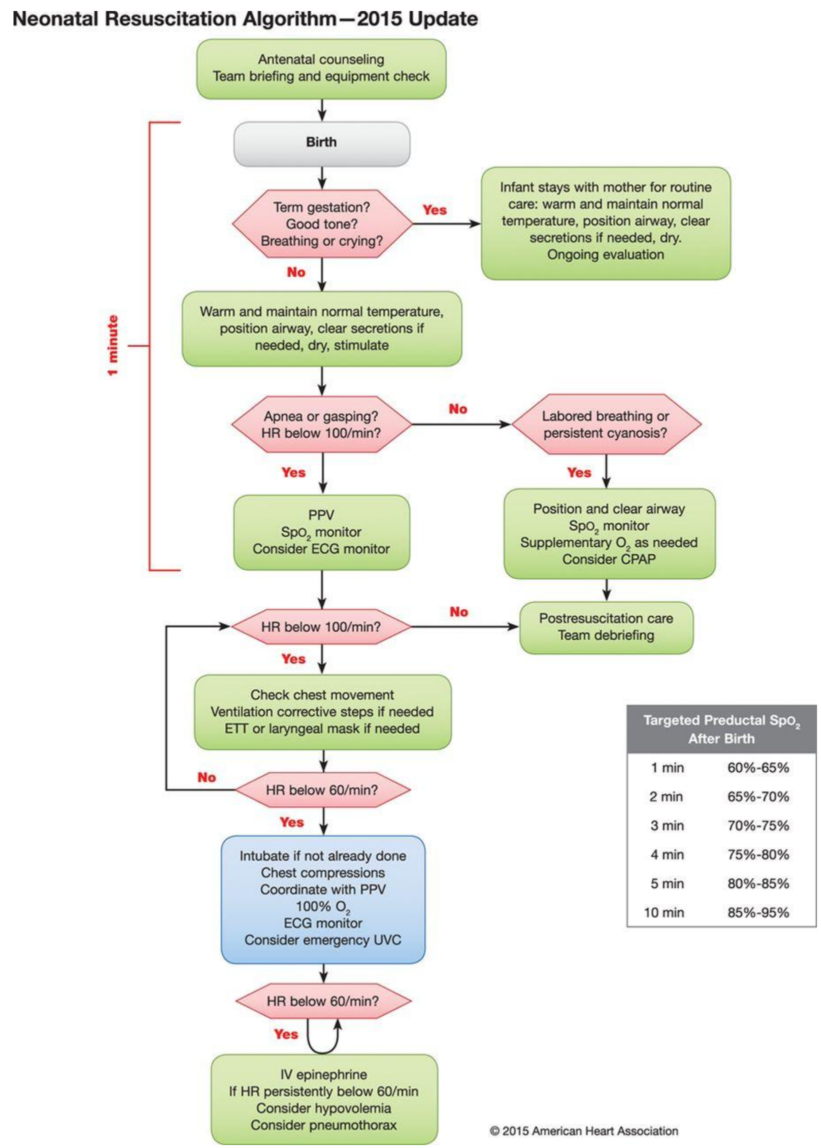
# Pediatric Cardiac Arrest Algorithm—2015 Update.




Allan R. de Caen et al. Circulation. 2015;132:S526-S542

American Heart Association.

# Neonatal Resuscitation Algorithm—2015 Update.




| Targeted Preductal $SpO_2$ After Birth | |
|---|---|
| 1 min | 60%-65% |
| 2 min | 65%-70% |
| 3 min | 70%-75% |
| 4 min | 75%-80% |
| 5 min | 80%-85% |
| 10 min | 85%-95% |

**Myra H. Wyckoff et al. Circulation. 2015;132:S543-S560**